TCGA-BRCA RNA-seq: Exploratory Analysis and Visualization

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1 Introduction

Studying the molecular basis of breast cancer helps reveal how tumors develop and respond to treatment. The Cancer Genome Atlas (TCGA) Breast Invasive Carcinoma (TCGA-BRCA) dataset provides RNA-seq count data and metadata from both tumor and normal samples, offering an opportunity to explore gene expression patterns across different patient groups [1]. In this project, a subset of this dataset was analyzed to visualize and interpret differences in gene expression using R. The goal was to understand how expression levels vary by sample type and identify trends that may reflect underlying biological patterns in breast cancer.

Two genes were selected for analysis: TSPAN6 (ENSG00000000003.15) and TNMD (ENSG00000000005.6). TSPAN6 is part of the tetraspanin family, which includes proteins that help regulate cell growth, movement, and communication[6]. TNMD encodes a protein that influences tissue growth and helps control the formation of blood vessels[7]. These genes were chosen to compare their expression between tumor and normal tissues and see how they might contribute to cancer-related cellular processes.

This report combines all analyses developed throughout the course, incorporating feedback from earlier check-ins. It includes summary statistics and visualizations such as histograms, scatter plots, violin plots, and a refined heatmap with annotation to highlight expression patterns among selected genes. An additional correlation matrix was also created to explore relationships between genes. Together, these visualizations show how computational tools can help uncover meaningful patterns in RNA-seq data and better understand gene activity in breast cancer.

2 Methods

The RNA-seq count matrix and associated metadata for breast cancer samples were obtained from the TCGA Breast Invasive Carcinoma (TCGA-BRCA) cohort [1]. Both files were imported into R and two genes, TSPAN6 and TNMD, were selected for initial analysis. All analyses were conducted in R (version 2025.09.0+387)[2]. Key packages included ggplot2 and scales for data visualization, dplyr for data manipulation, ComplexHeatmap and circlize for multi-gene expression visualization, and corrplot for correlation analysis [3, 4, 5].

Two genes from the count matrix were initially selected for exploratory analysis. For each gene, summary statistics–including mean, median, standard deviation, minimum, and maximum–were computed to describe the variability of expression between samples. A histogram was generated to visualize the distribution of counts for the primary gene, TSPAN6. To assess the expression between primary and secondary gene, TNMD, a log-transformed scatter plot with a fitted linear regression line was created between the two selected genes. A violin plot was then produced to compare expression of the main gene across the sample type covariate–Primary Tumor, Solid Tissue Normal, Metastatic–using a custom color palette and overlaid boxplots for clarity.

Ten genes were then randomly selected to generate a heatmap summarizing expression patterns across samples, annotated by sample type to highlight clustering and variation among groups. To address feedback from prior check-ins, expression counts were log-transformed to reduce skew, and a more contrasting color palette was applied for improved readability. Finally, a correlation matrix was constructed to visualize pairwise relationships among the ten selected genes. Before calculating correlations, genes with zero variance were removed to avoid computational errors. Pearson correlation coefficients were computed using log-transformed data, and the resulting matrix was visualized using hierarchical clustering and a red–white–blue diverging color scale to emphasize the strength and direction of gene–gene relationships.

3 Results

3.1 Summary Statistics

The summary statistics in Table 1 describe the overall distribution of expression counts for the selected gene, TSPAN6, across all breast cancer samples. The mean count value was approximately 3,208, with a median of 2,700, indicating a right-skewed distribution influenced by multiple high-expression samples. The wide range, from a minimum of 69 to a maximum of 40,780, and a standard deviation above 2,500 show substantial variability in expression levels among samples. This variation suggests differences in gene activity across patient samples, reflecting biological diversity within the TCGA-BRCA cohort.

Statistic	Value
Mean	3208.132
Median	2700
Standard Deviation	2510.336
Minimum	69
Maximum	40780

Table 1: Summary of Statistics for Distribution of TSPAN6 Gene Expression Counts

3.2 Histogram

The histogram in Figure 1 displays the distribution of expression counts for the gene TSPAN6 across all samples in the TCGA-BRCA dataset. The majority of samples show low expression levels, with a sharp peak near the lower count range, indicating that this gene is expressed at modest levels in most tissues. However, a small number of samples show extremely high expression values, creating a long right tail and confirming the positive skew observed in the summary statistics. This skew indicates that certain tumor samples have increased gene expression compared to normal tissues.

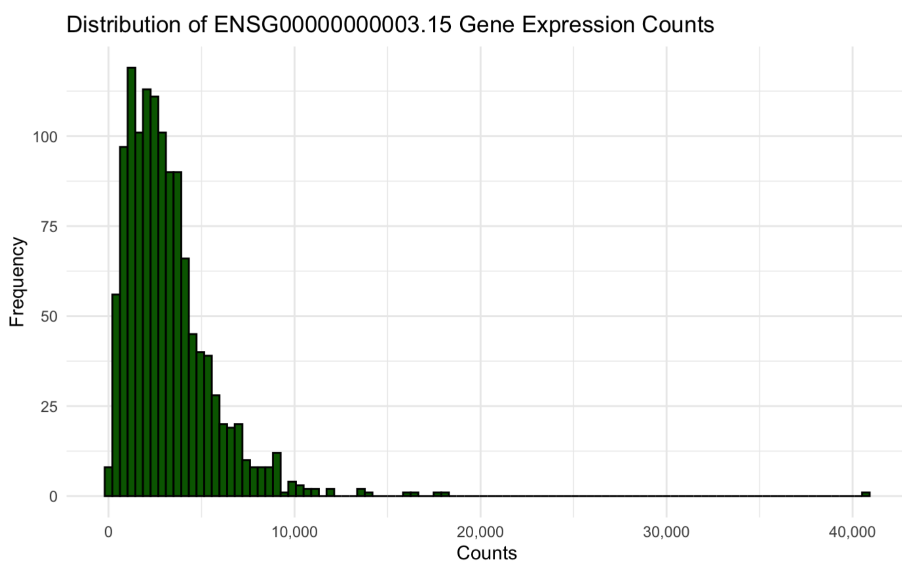

Figure 1: Histogram of Distribution of TSPAN6 Gene Expression Counts

3.3 Scatter Plot

The scatter plot in Figure 2 compares the log-transformed expression counts of TSPAN6 and TNMD across all samples. Each point represents an individual sample, and the fitted linear regression line (y = 0.0314x + 37.82) indicates a small positive relationship between the two genes. This suggests that as the expression of TSPAN6 increases, the expression of TNMD also tends to increase, although with considerable variability between samples. The broad spread of points around the regression line highlights variation in gene expression within the TCGA-BRCA cohort and may reflect differences in how this gene is regulated across tumor subtypes.

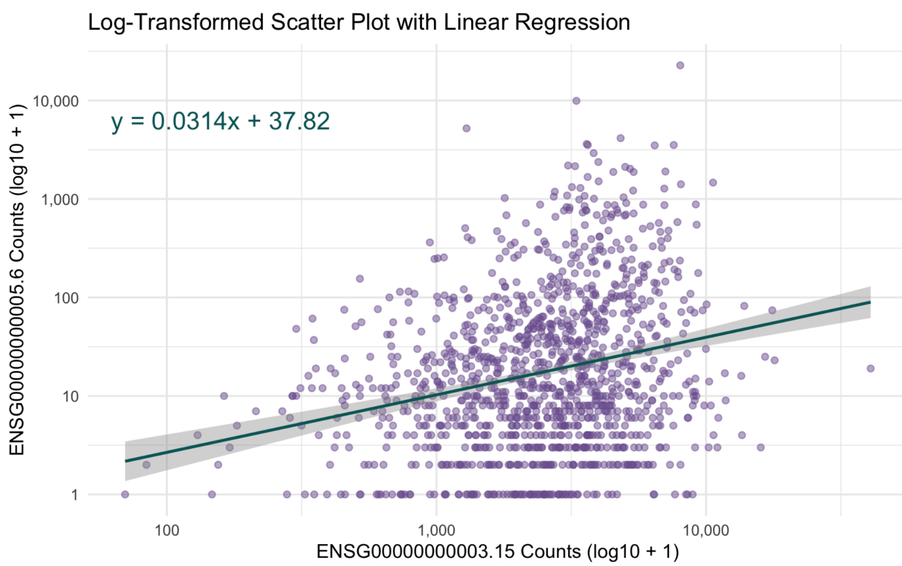

Figure 2: Log-Transformed Scatter Plot with Linear Regression Between Two Genes

3.4 Violin Plot

The violin plot in Figure 3 illustrates the distribution of TSPAN6 expression counts across different sample types in the TCGA-BRCA dataset: Metastatic, Primary Tumor, and Solid Tissue Normal. Each violin displays the density and variability of expression within each group, with an overlaid boxplot highlighting the median and interquartile range. Expression levels are higher and more variable in metastatic and primary tumor samples than in normal tissues which suggests greater gene activity in cancer cells. The wider range of values in tumor samples also reflects the biological diversity seen in breast cancer progression and metastasis.

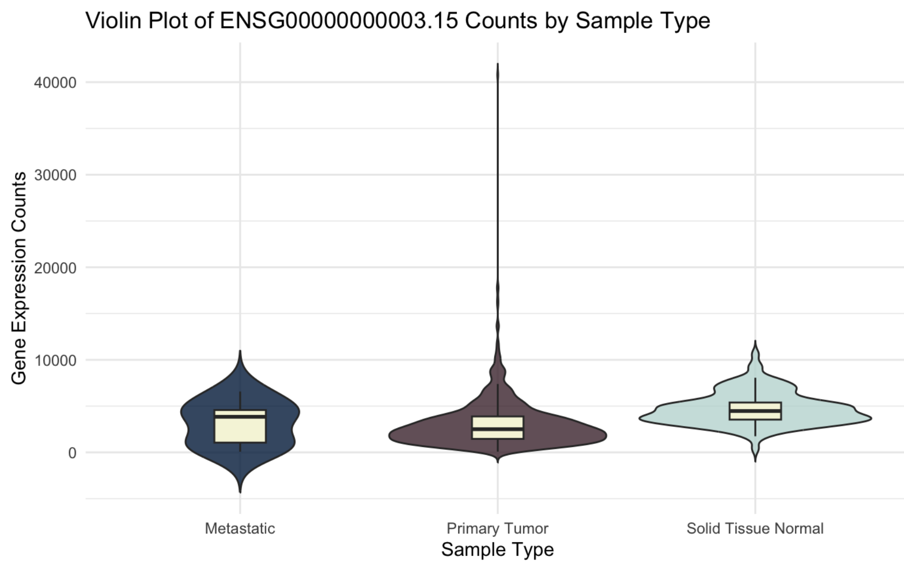

Figure 3: Violin Plot of TSPAN6 Counts by Sample Type

3.5 Heatmap

The heatmap in Figure 4 visualizes log-transformed expression levels for ten randomly selected genes across all TCGA-BRCA samples, grouped by sample type. Each column represents a sample and each row a gene, with expression intensity indicated by color gradients from dark blue (low expression) to magenta (high expression). The annotation bar at the top distinguishes the sample categories–Primary Tumor, Normal Solid Tissue, and Metastatic–allowing direct comparison between biological conditions. Distinct clusters are visible, with tumor samples showing higher and more variable expression than normal tissues. These patterns highlight differences in gene activity among breast cancer types and show how log scaling and color contrast help clarify complex data.

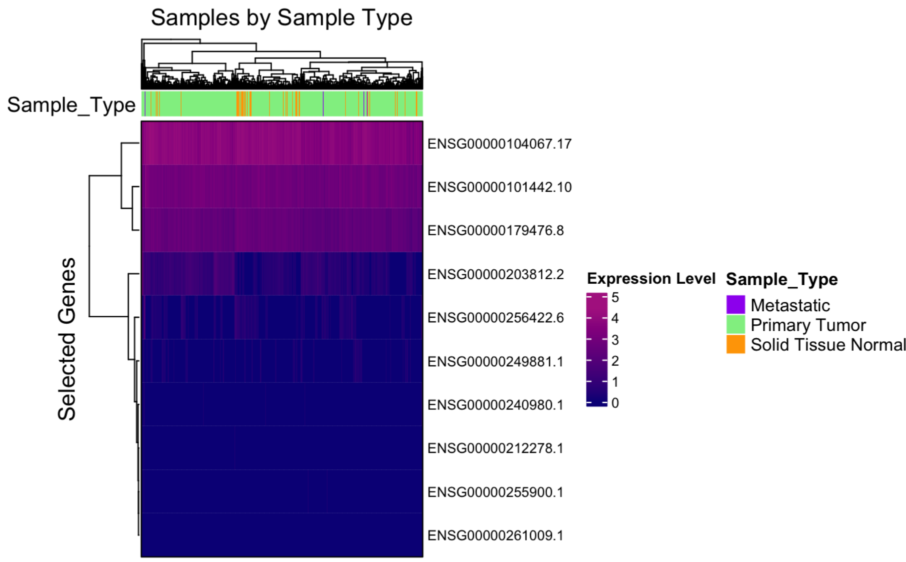

Figure 4: Samples by Sample Type

3.6 Additional Plot: Correlation Matrix

The correlation matrix in Figure 5 summarizes pairwise Pearson correlations among the ten genes previously analyzed in the heatmap. Each cell represents the strength and direction of the linear relationship between two genes, with darker blue shades indicating stronger positive correlations and lighter or reddish hues reflecting weaker or negative associations. Most genes are weakly to moderately correlated, suggesting they share some expression patterns but are not strongly linked. The weak correlations show that these genes vary widely in their expression within the TCGA-BRCA cohort, suggesting that while some may act in related pathways, others function independently.

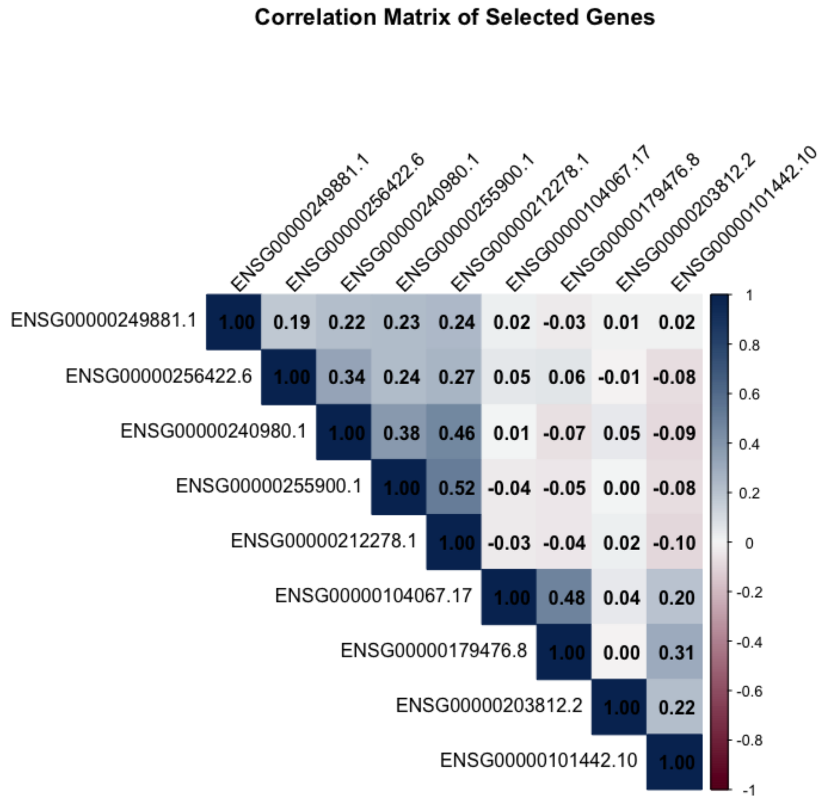

Figure 5: Correlation Matrix of Selected Ten Genes

4 Conclusion

Overall, this analysis demonstrates how visualization techniques can reveal meaningful patterns in complex RNA-seq data. The results highlight clear differences in gene expression between tumor and normal samples, as well as variation among genes that may reflect diverse biological functions within breast cancer. By combining multiple plots–including summary statistics, scatter, violin, heatmap, and correlation matrix–this project provides a concise but comprehensive view of expression trends in the TCGA-BRCA dataset[1].

References

[1] National Cancer Institute & National Human Genome Research Institute. (2025). *TCGA-BRCA: The Cancer Genome Atlas Breast Invasive Carcinoma Study*. Retrieved from `https://portal.gdc.cancer.gov/projects/TCGA-BRCA`

[2] R Core Team. (2025). *R: A language and environment for statistical computing.* R Foundation for Statistical Computing, Vienna, Austria.

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[4] Gu, Z. (2016). ComplexHeatmap: An R package for visualization of complex data. *Bioinformatics, 32*(18), 2847–2849.

[5] Wei, T., & Simko, V. (2021). *R package 'corrplot': Visualization of a correlation matrix* (Version 0.92). Retrieved from `https://github.com/taiyun/corrplot`

[6] Zhang Lab. (2023). *Gene: TSPAN6 – LinkedOmics Knowledge Base*. Retrieved from `https://kb.linkedomics.org/gene/TSPAN6`

[7] Zhang Lab. (2023). *Gene: TNMD – LinkedOmics Knowledge Base*. Retrieved from `https://kb.linkedomics.org/gene/TNMD`

Project Repository: All analysis code, R Markdown files, and figures are available at: `https://github.com/camitor/TCGA-BRCA-FinalProject`